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# SMARTPHONE SPECIFICATION BY BRANDS

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SEPTEMBER 6, 2024  
IS 1006 FUNDAMENTALS OF STATISTICS  
Group 10

## Table of Content

|                                                      |    |
|------------------------------------------------------|----|
| 1. Table of Figures .....                            | 3  |
| 2. Introduction.....                                 | 5  |
| ➤ Objectives of the study.....                       | 6  |
| 3. Methodology .....                                 | 6  |
| ➤ Types of charts .....                              | 7  |
| ➤ Numerical summaries .....                          | 10 |
| ➤ Correlation and Regression.....                    | 11 |
| ➤ Hypothesis.....                                    | 13 |
| 4. Data .....                                        | 14 |
| ➤ Qualitative variables (categorical variables)..... | 15 |
| ➤ Quantitative variables .....                       | 15 |
| 5. Analysis.....                                     | 16 |
| ➤ Univariate analysis.....                           | 16 |
| ▪ Ratings .....                                      | 16 |
| ▪ Ratings of brands .....                            | 18 |
| ▪ Number of brands .....                             | 19 |
| ▪ Selling price .....                                | 20 |
| ▪ Original Price .....                               | 23 |
| ▪ Storage .....                                      | 25 |
| ▪ Memory .....                                       | 26 |

|                                          |    |
|------------------------------------------|----|
| ➤ Bivariate analysis .....               | 27 |
| ▪ Original price and selling price ..... | 28 |
| ▪ Selling price and Brand .....          | 30 |
| ▪ Rating and Brand .....                 | 31 |
| ▪ Original Price and Brand .....         | 32 |
| 6. Correlation .....                     | 34 |
| ▪ Selling Price and Original Price ..... | 34 |
| ▪ Ratings and selling price .....        | 35 |
| ▪ Ratings and Original Price .....       | 36 |
| 7. Regression .....                      | 38 |
| ▪ Selling Price and Original Price ..... | 38 |
| ▪ Original Price and Rating .....        | 41 |
| 8. Discussion .....                      | 45 |
| 9. Conclusion .....                      | 46 |
| 10. References .....                     | 47 |
| 11. Group Members .....                  | 47 |

## Table of Figures

|                                                              |    |
|--------------------------------------------------------------|----|
| Figure 1 intrdouction .....                                  | 5  |
| Figure 2 Excel and Minitab .....                             | 6  |
| Figure 3 Pie Chart .....                                     | 7  |
| Figure 4 Box Plot.....                                       | 8  |
| Figure 5 Histogram .....                                     | 8  |
| Figure 6 Bar chart .....                                     | 9  |
| Figure 7 Scatter Plot .....                                  | 9  |
| Figure 8 Numerical Summaries_1 .....                         | 10 |
| Figure 9 Numerical Summaries_2 .....                         | 11 |
| Figure 10 Correlation Vs Regression .....                    | 12 |
| Figure 11 Hypothesis .....                                   | 13 |
| Figure 12 Data.....                                          | 14 |
| Figure 13 Histogram of Rating .....                          | 16 |
| Figure 14 Descriptive Analysis of Rating .....               | 17 |
| Figure 15 Boxplot of Rating.....                             | 17 |
| Figure 16 Pie chart of Brands.....                           | 18 |
| Figure 17 Bar chart of Brands .....                          | 19 |
| Figure 18 Histogram of Selling Price.....                    | 20 |
| Figure 19 Descriptive Analysis of selling Price .....        | 20 |
| Figure 20 Boxplot of Selling Price .....                     | 21 |
| Figure 21 Boxplot of selling Price with reduce scale .....   | 22 |
| Figure 22 Histogram of Original Price .....                  | 23 |
| Figure 23 Descriptive Analysis of Original Price .....       | 23 |
| Figure 24 Boxplot of Original Price .....                    | 24 |
| Figure 25 Boxplot with original price with reduce scale..... | 25 |
| Figure 26 Bar chart of storage.....                          | 25 |
| Figure 27 Bar chart of Memory .....                          | 26 |
| Figure 28 Boxplot of Original Price and Selling Price .....  | 28 |

|                                                                                  |    |
|----------------------------------------------------------------------------------|----|
| Figure 29 Descriptive Analysis of Original price and selling price.....          | 28 |
| Figure 30 Scatter Plot of Original Price and selling Price .....                 | 29 |
| Figure 31 Dot plot of selling price and Brand.....                               | 30 |
| Figure 32 Dot plot of Rating and Brand .....                                     | 31 |
| Figure 33 Dot plot of Original Price and Brand .....                             | 32 |
| Figure 34 Matrix plot of Selling Price and Original Price.....                   | 34 |
| Figure 35 Correlation of Selling price Vs Original Price .....                   | 34 |
| Figure 36 Matrix plot of Rating and selling Price .....                          | 35 |
| Figure 37 Correlation of Rating Vs Selling Price .....                           | 35 |
| Figure 38 Matrix plot of Rating and Original Price .....                         | 36 |
| Figure 39 Correlation of Rating Vs Original Price .....                          | 36 |
| Figure 40 Plot of Original Price Vs selling Price with Fitted Line .....         | 38 |
| Figure 41 Numerical Measures of selling price Vs original price fitted line..... | 38 |
| Figure 42 Plot of Rating Vs Original Price with fitted line.....                 | 41 |
| Figure 43 Numerical Measures of Original price Vs Rating fitted line .....       | 42 |
| Figure 44 Conclusion.....                                                        | 46 |

## Introduction



*Figure 1introduction*

The smartphone market is highly competitive with various brands offering distinct specs and innovations. Understanding the specification and unique selling points of different brands can provide valuable insights into market trends and consumer preferences. This project aims to analyze the specifications of smartphones by major brands to identify patterns, selling price and strength.

Understanding the factors that drive mobile sales is crucial for manufacturers and retailers to devise effective strategies and maximize revenue. We aim to deliver a robust tool that enhances understanding, supports better purchasing decisions, and contributes to a more informed and connected technology community.

## Objectives of the study

1. **Compare and contrast** the technical specifications of leading smartphone brands.
2. **Identify unique selling points** and differentiating specs of each brand.
3. **Analyze market trends** and consumer preferences based on specifications.

## Methodology



*Figure 2 Excel and Minitab*

Methods used to find out the behavior of collected data and organize them into an effective manner are very important. Here statistical software called Minitab and Microsoft excel are used. Analysis is carried out by using graphical methods such as bar charts, pie charts, boxplot and histograms. Also mean median, standard deviations and quartiles are some numerical statistical summaries used here.

## Types of charts

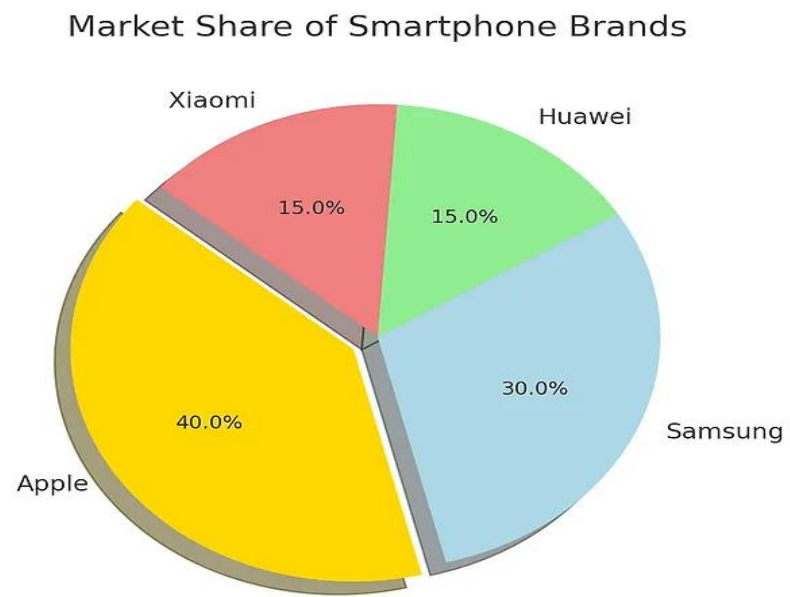


Figure 3 Pie Chart



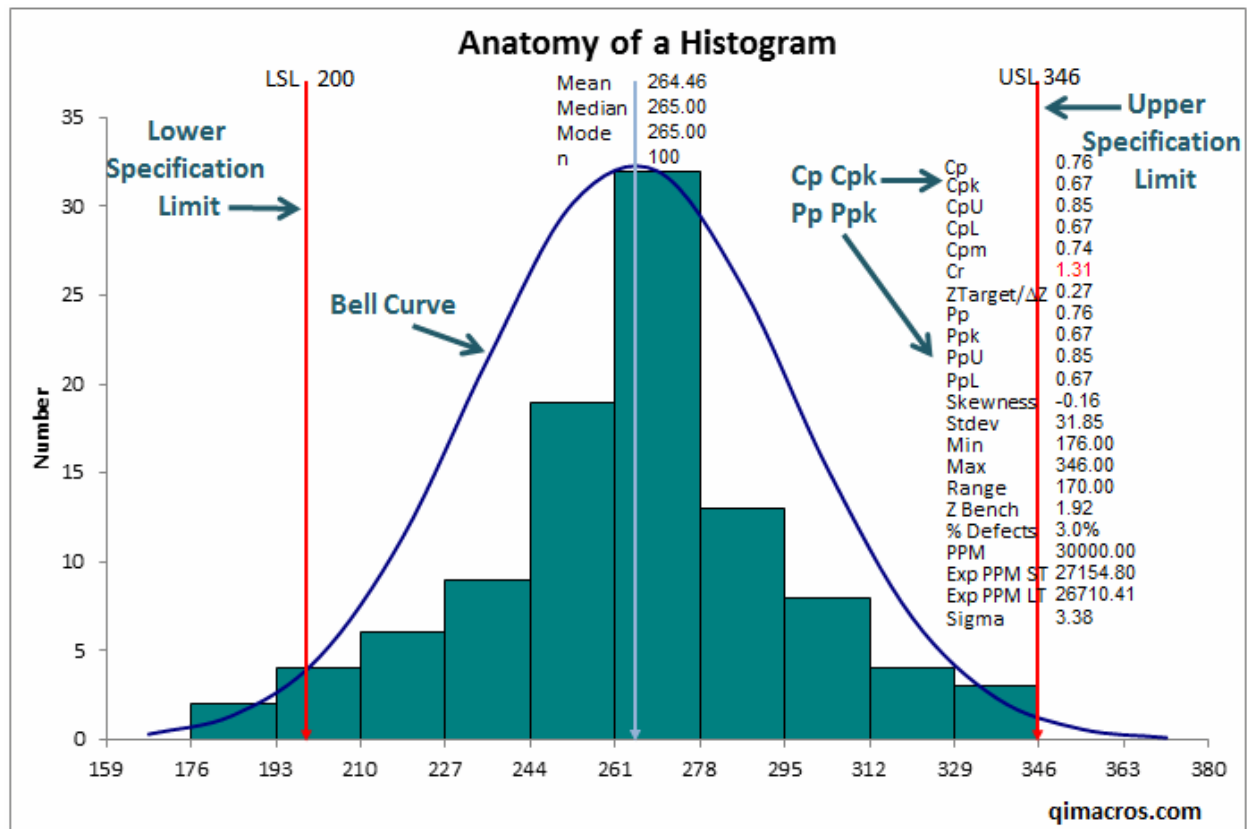


Figure 5 Histogram

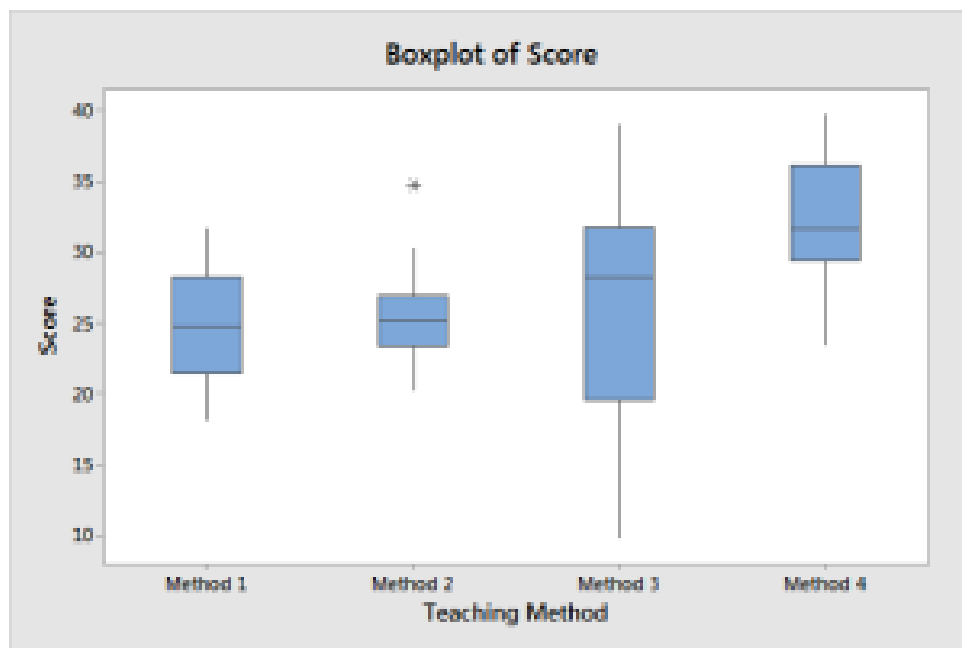


Figure 4 Box Plot

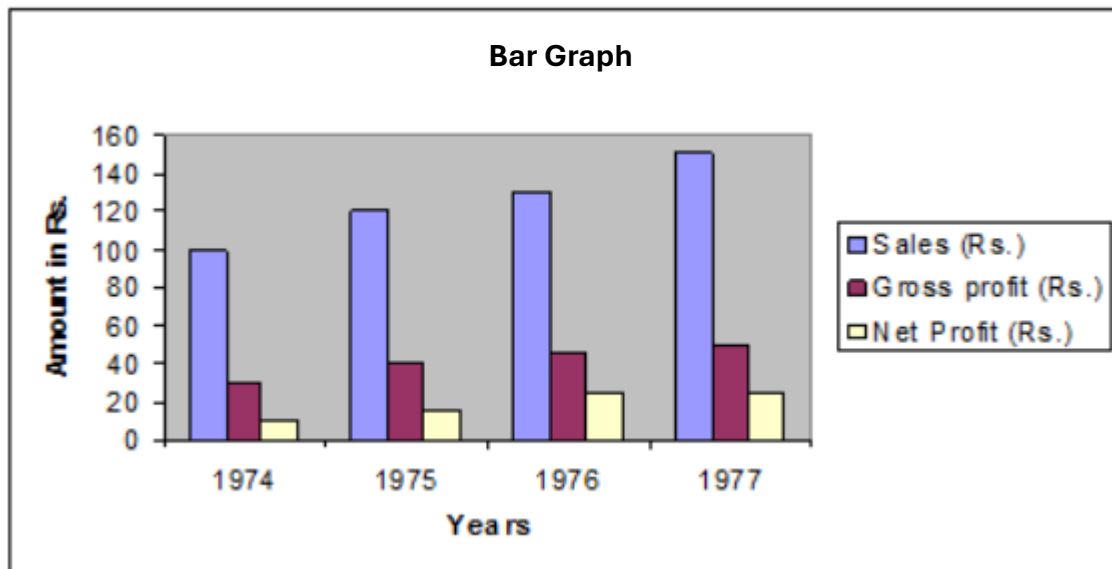


Figure 6 Bar chart

## Scatter Diagram

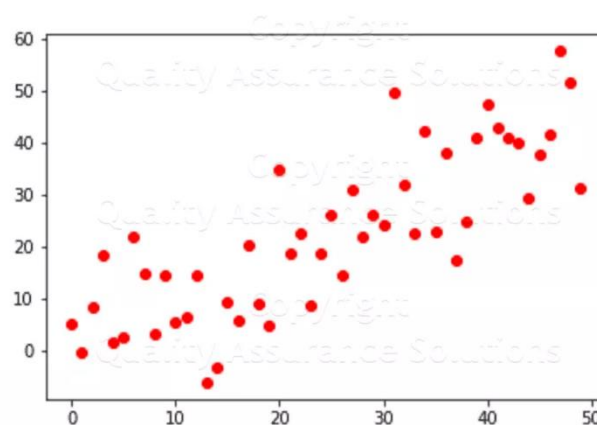


Figure 7 Scatter Plot

## Numerical summaries

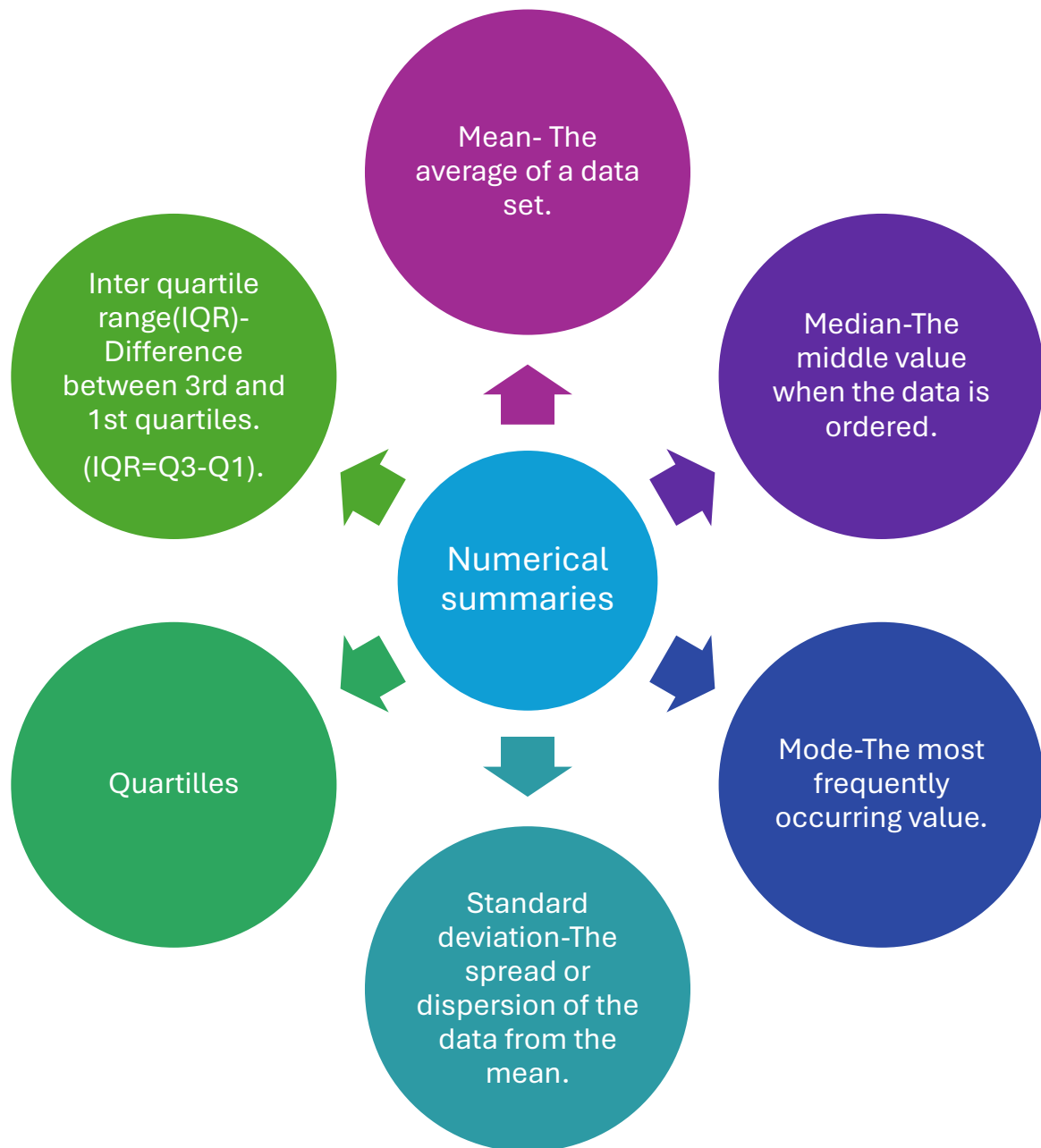


Figure 8 Numerical Summaries\_1

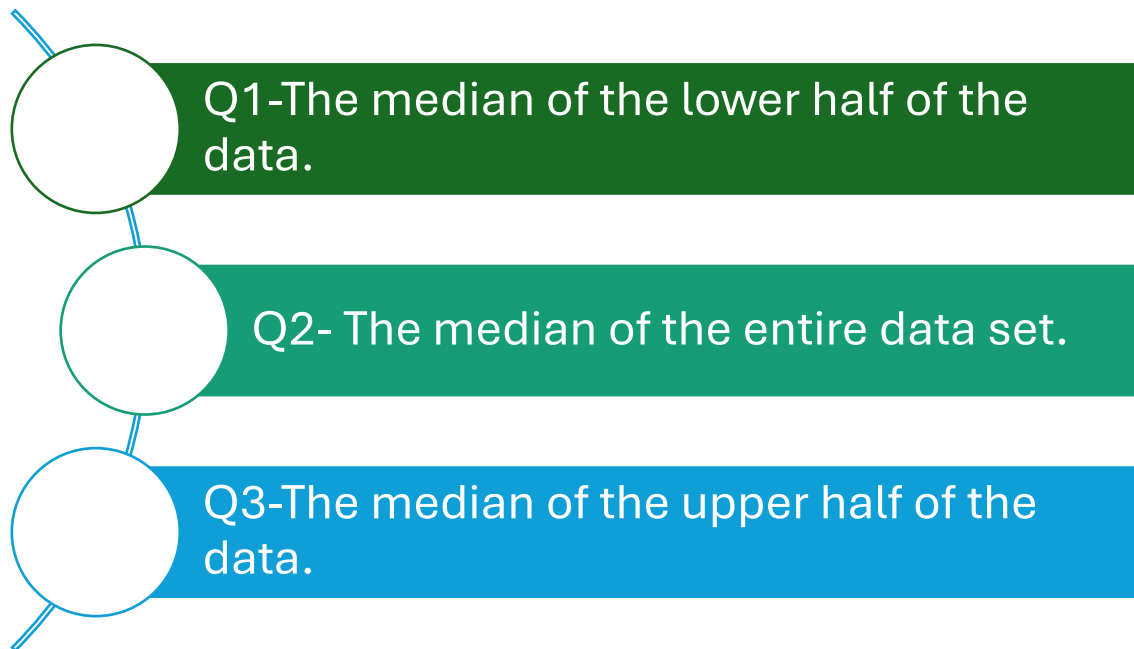
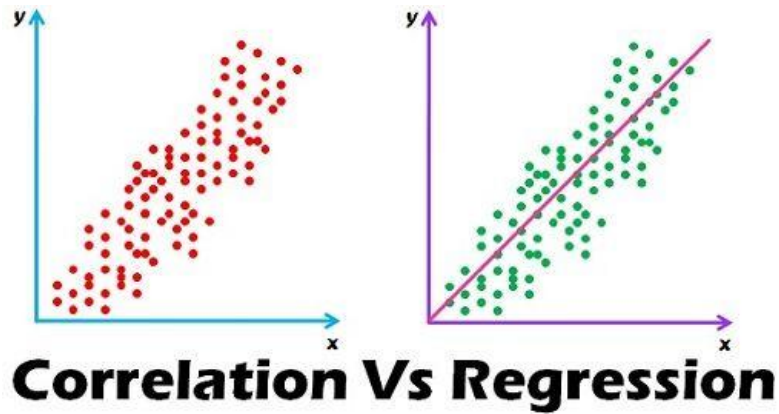


Figure 9 Numerical Summaries\_2

## Correlation and Regression

- ❖ **Correlation** - Measures the direction and strength of the linear association between two quantitative variables.
- ❖ **Regression** - Expresses the relationship between two variables in the form of an equation.
- ❖ **Regression Line** - How a response variable  $y$  changes as an explanatory variable  $x$  change.



### Differences Between Correlation and Regression

| Correlation                                                                                             | Regression                                                                                      |
|---------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| 1 Tests for a linear relationship between two variables                                                 | Determines the impact of a change in one variable on the other                                  |
| 2 Generally symmetrical: the correlation between X and Y is the same as the correlation between Y and X | The regression of X on Y is not the same as the regression of Y on X                            |
| 3 Generally not used for predictions                                                                    | Often used for predictions                                                                      |
| 4 Order of variables does not matter                                                                    | Order of variables is important; the independent and dependent variables cannot be interchanged |
| 5 Is limited to two variables                                                                           | More than one independent variable is possible (i.e., multiple regression)                      |

Figure 10 Correlation Vs Regression

## Hypothesis

A hypothesis is a testable statement or prediction about the relationship between two or more variables, often formulated to be tested through research and experimentation.

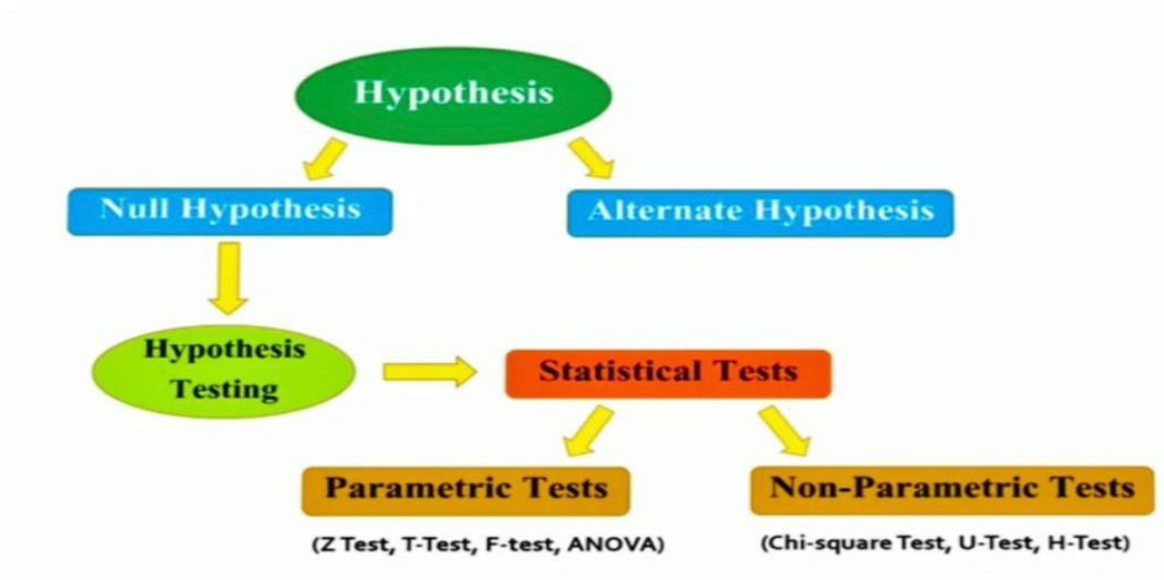


Figure 11 Hypothesis

## Data

|    | C1-T<br>Brand | C2-T<br>Model | C3-T<br>Color    | C4-T<br>Memory | C5-T<br>Storage | C6<br>Rating | C7<br>Selling Price | C8<br>Original Price | C9 | C10 | C11 | C12 | C13 |
|----|---------------|---------------|------------------|----------------|-----------------|--------------|---------------------|----------------------|----|-----|-----|-----|-----|
| 1  | OPPO          | A53           | Moonlight Black  | 4 GB           | 64 GB           | 4.5          | 11990               | 15990                |    |     |     |     |     |
| 2  | OPPO          | A53           | Mint Cream       | 4 GB           | 64 GB           | 4.5          | 11990               | 15990                |    |     |     |     |     |
| 3  | OPPO          | A53           | Moonlight Black  | 6 GB           | 128 GB          | 4.3          | 13990               | 17990                |    |     |     |     |     |
| 4  | OPPO          | A53           | Mint Cream       | 6 GB           | 128 GB          | 4.3          | 13990               | 17990                |    |     |     |     |     |
| 5  | OPPO          | A53           | Electric Black   | 4 GB           | 64 GB           | 4.5          | 11990               | 15990                |    |     |     |     |     |
| 6  | OPPO          | A53           | Electric Black   | 6 GB           | 128 GB          | 4.3          | 13990               | 17990                |    |     |     |     |     |
| 7  | OPPO          | A12           | Deep Blue        | 4 GB           | 64 GB           | 4.4          | 10490               | 11990                |    |     |     |     |     |
| 8  | OPPO          | A12           | Black            | 3 GB           | 32 GB           | 4.4          | 9490                | 10990                |    |     |     |     |     |
| 9  | OPPO          | A12           | Blue             | 3 GB           | 32 GB           | 4.4          | 9490                | 10990                |    |     |     |     |     |
| 10 | OPPO          | A12           | Flowing Silver   | 3 GB           | 32 GB           | 4.4          | 9490                | 10990                |    |     |     |     |     |
| 11 | OPPO          | A12           | Deep Blue        | 3 GB           | 32 GB           | 4.4          | 9490                | 10990                |    |     |     |     |     |
| 12 | OPPO          | A12           | Flowing Silver   | 4 GB           | 64 GB           | 4.4          | 10490               | 11990                |    |     |     |     |     |
| 13 | OPPO          | A53s 5G       | Crystal Blue     | 6 GB           | 128 GB          | 4.3          | 15990               | 16990                |    |     |     |     |     |
| 14 | OPPO          | A53s 5G       | Ink Black        | 6 GB           | 128 GB          | 4.3          | 15990               | 16990                |    |     |     |     |     |
| 15 | OPPO          | A12           | Blue             | 4 GB           | 64 GB           | 4.4          | 10490               | 11990                |    |     |     |     |     |
| 16 | OPPO          | A53s 5G       | Crystal Blue     | 8 GB           | 128 GB          | 4.3          | 17990               | 18990                |    |     |     |     |     |
| 17 | OPPO          | A53s 5G       | Ink Black        | 8 GB           | 128 GB          | 4.3          | 17990               | 18990                |    |     |     |     |     |
| 18 | OPPO          | A33           | Moonlight Black  | 3 GB           | 32 GB           | 4.3          | 10490               | 12990                |    |     |     |     |     |
| 19 | OPPO          | A31           | Lake Green       | 4 GB           | 64 GB           | 4.3          | 11960               | 12990                |    |     |     |     |     |
| 20 | OPPO          | A31           | Mystery Black    | 4 GB           | 64 GB           | 4.3          | 11779               | 11919                |    |     |     |     |     |
| 21 | OPPO          | A74 5G        | Fantastic Purple | 6 GB           | 128 GB          | 4.3          | 18979               | 18979                |    |     |     |     |     |
| 22 | OPPO          | A11K          | Flowing Silver   | 2 GB           | 32 GB           | 4.2          | 8990                | 10990                |    |     |     |     |     |
| 23 | OPPO          | F17 Pro       | Matte Black      | 8 GB           | 128 GB          | 4.3          | 19990               | 25990                |    |     |     |     |     |
| 24 | OPPO          | A54           | Moonlight Gold   | 4 GB           | 128 GB          | 4.3          | 15990               | 15990                |    |     |     |     |     |
| 25 | OPPO          | A54           | Moonlight Gold   | 4 GB           | 64 GB           | 4.3          | 14990               | 14990                |    |     |     |     |     |
| 26 | OPPO          | Reno6 5G      | Aurora           | 8 GB           | 128 GB          | 4.3          | 29990               | 35990                |    |     |     |     |     |
| 27 | OPPO          | Reno6 5G      | Stellar Black    | 8 GB           | 128 GB          | 4.3          | 29990               | 35990                |    |     |     |     |     |
| 28 | OPPO          | F17 Pro       | Magic Blue       | 8 GB           | 128 GB          | 4.3          | 19990               | 25990                |    |     |     |     |     |
| 29 | OPPO          | A54           | Starry Blue      | 4 GB           | 64 GB           | 4.3          | 14990               | 14990                |    |     |     |     |     |

Figure 12 Data

This data set is abstracted from a well-known web site called Kaggle which is an online machine learning environment and community. It has standard datasets that hundreds or thousands of individuals or teams try to model, and there is a leaderboard for each competition.

<https://www.kaggle.com/datasets/devsubhash/flipkart-mobiles-dataset>

Selected data set contains **2647 observations** and the following variables,

## **Qualitative variables (categorical variables)**

Categorical data represents characteristics or attributes that can be divided into distinct groups or categories, often described using labels or names, without any inherent numerical value or order.

1. Brand
2. Model
3. Color
4. Memory
5. Storage

## **Quantitative variables**

Quantitative variables are numerical values that represent measurable quantities, allowing for arithmetic operations and statistical analysis, and can be either discrete (countable) or continuous (measurable).

1. Rating
2. Selling price
3. Original price

We are hoping to analyze these qualitative and quantitative variables according to our main objectives.



## Analysis

### Univariate analysis

Univariate analysis involves the examination and analysis of a single variable, focusing on its distribution, central tendency, and dispersion to summarize and understand its characteristics.

### Ratings

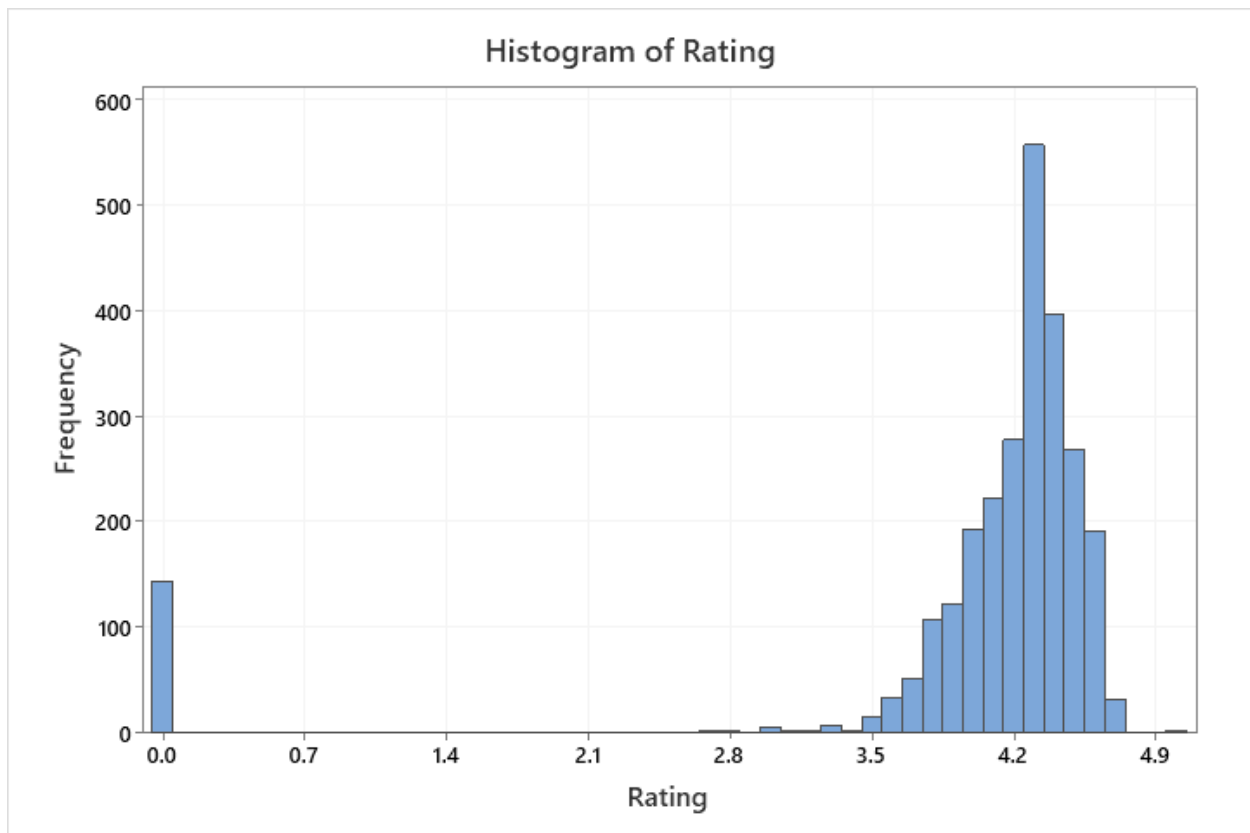


Figure 13 Histogram of Rating

## Statistics

| Variable | N    | N* | Mean   | SE Mean | StDev  | Minimum | Q1     | Median | Q3     | Maximum |
|----------|------|----|--------|---------|--------|---------|--------|--------|--------|---------|
| Rating   | 2647 | 0  | 4.0017 | 0.0194  | 0.9975 | 0.0000  | 4.0000 | 4.3000 | 4.4000 | 5.0000  |

Figure 14 Descriptive Analysis of Rating

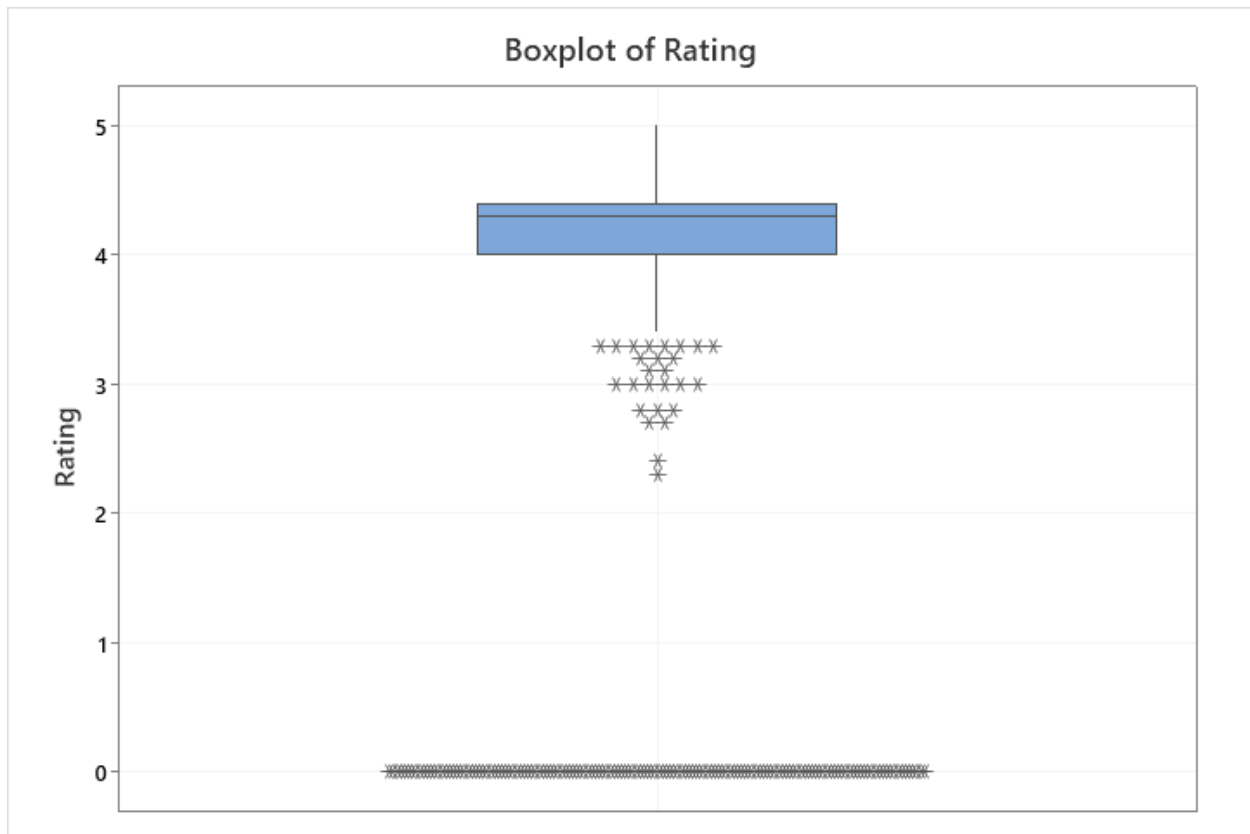


Figure 15 Boxplot of Rating

We can see this Distribution is approximately Negatively skewed. And it is Unimodal. We can see only one peak.

And There is an outlier. (According to IQR \*1.5 rule).

- The data is centered around ratings 4.3.
- These all brands have ratings between 0.6-5.0 ratings.

- Most brands rating is 4.3.
- We can see what the maximum and minimum ratings in this data are by looking above histogram.

## Ratings of brands

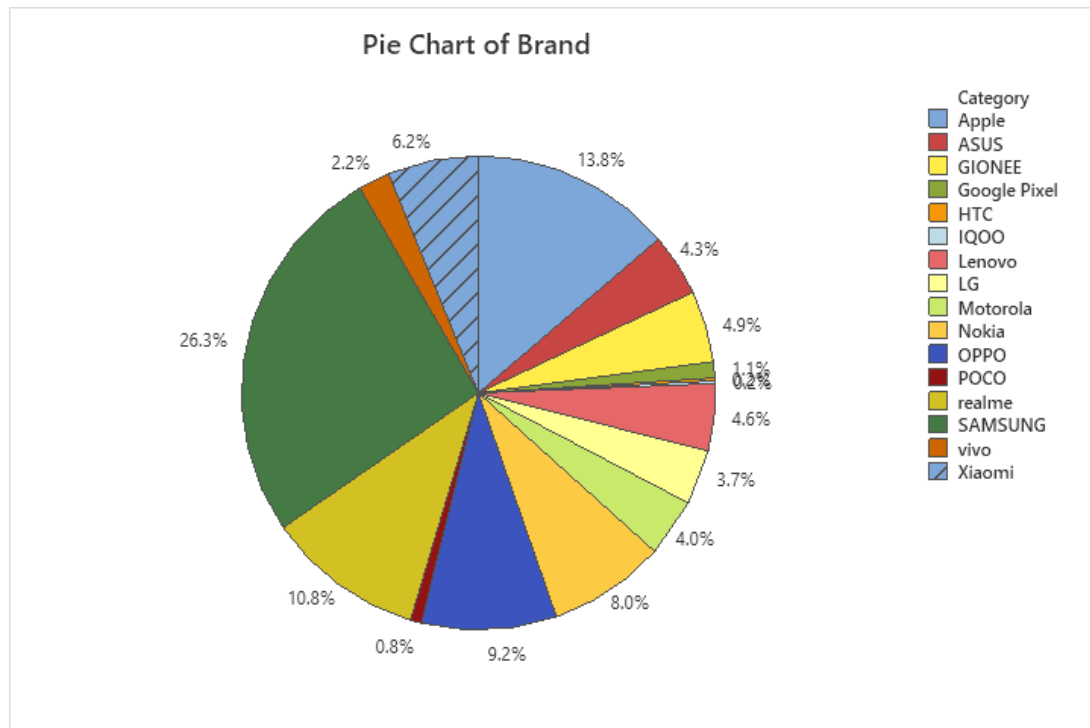


Figure 16 Pie chart of Brands

- The pie chart represents the distribution of various smartphone brands ratings.
- It represents 16 brands.
- Samsung holds the largest share at 26.3%.
- Brands like HTC, Poco, and IQOO have smaller portions.

## Number of brands

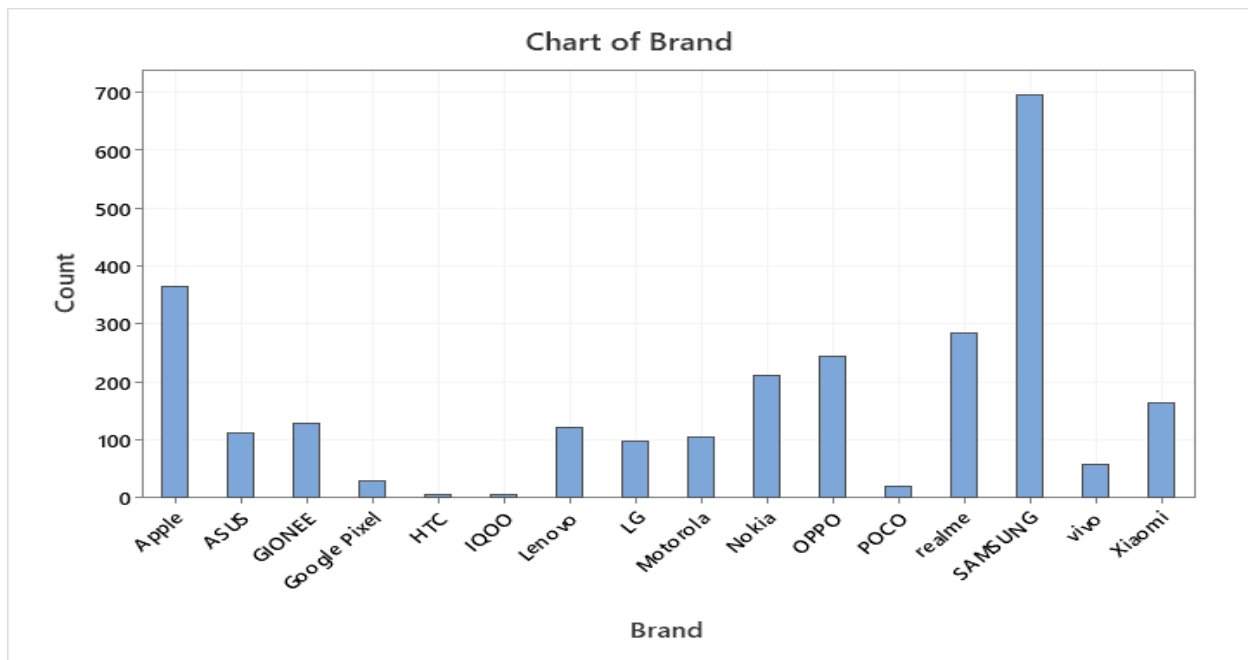


Figure 17 Bar chart of Brands

This chart also indicates univariate analysis of Number of smartphone brands. According to this bar chart we can see Most People use Samsung phone brand. (695, 26.3%) and the least people use HTC and IQOO phone brands (5, 0.18%).

## Selling price

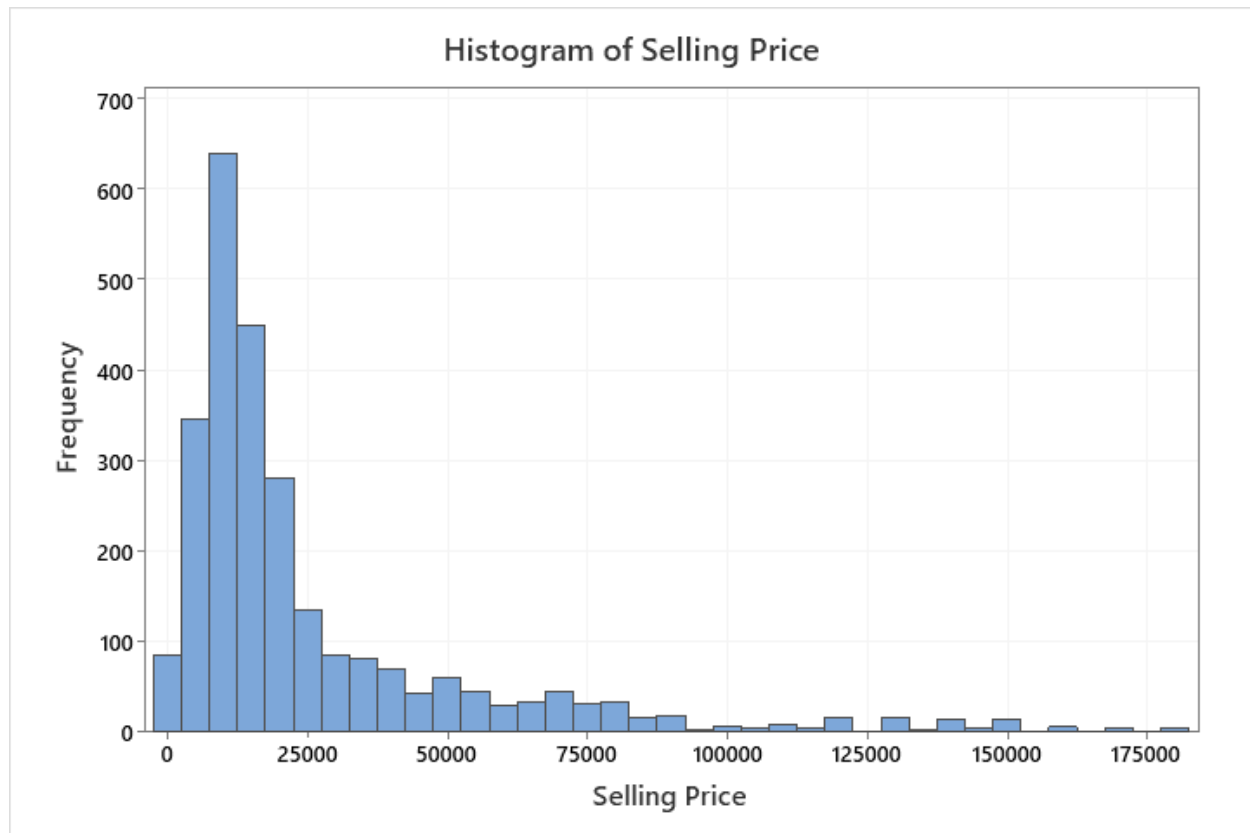


Figure 18 Histogram of Selling Price

We can see this Distribution is approximately positively skewed. And it is Unimodal. We can see only one peak.

## Statistics

| Variable      | N    | N* | Mean  | SE Mean | StDev | Minimum | Q1   | Median | Q3    | Maximum |
|---------------|------|----|-------|---------|-------|---------|------|--------|-------|---------|
| Selling Price | 2644 | 3  | 26461 | 580     | 29833 | 1000    | 9490 | 14999  | 29999 | 179900  |

Figure 19 Descriptive Analysis of selling Price

And There are outliers. (According to IQR \*1.5 rule).

- The data is centered around selling price RS.14999.

- These all selling price between 0 – 179900 rupees.
- Most brands have selling prices between 7500 – 12500 rupees.
- We can see what the maximum and minimum selling prices are by looking above bar chart.

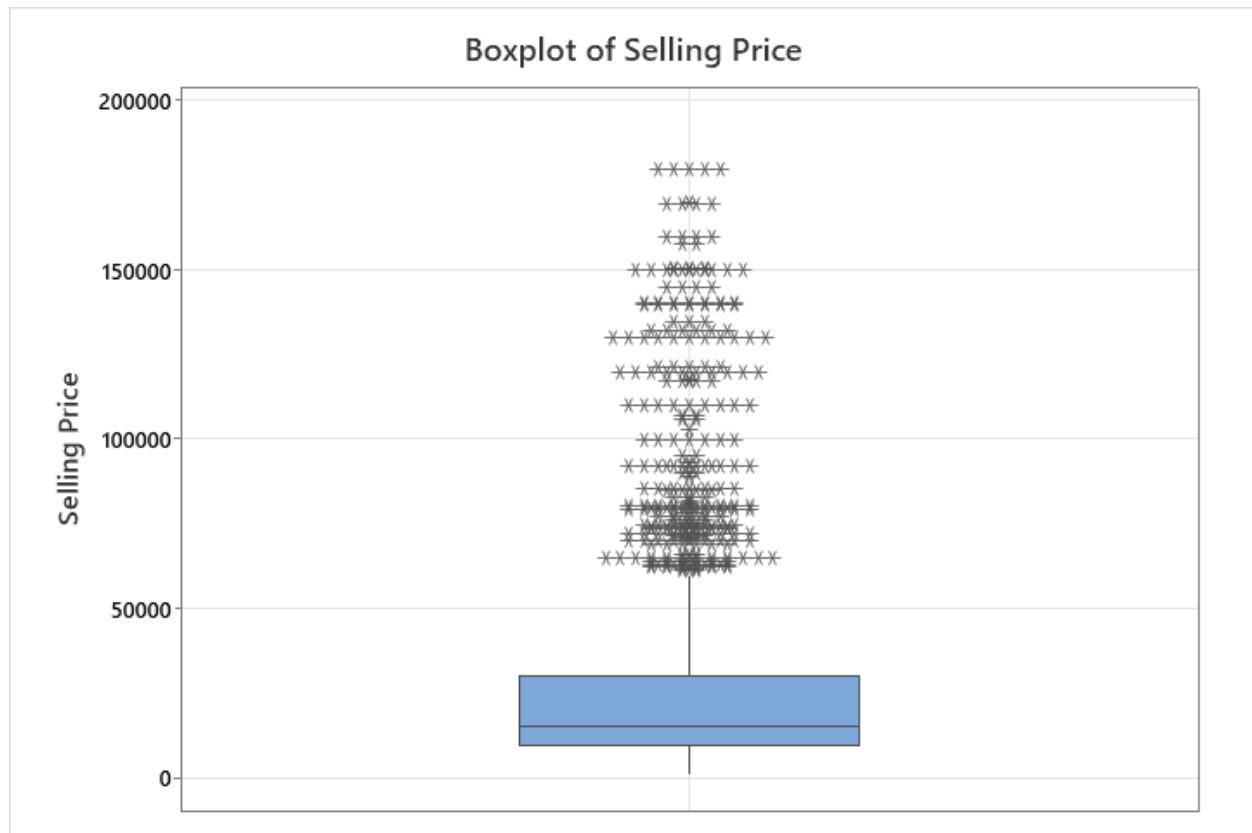


Figure 20 Boxplot of Selling Price

As we can see there are lot of outliers for the box plot of selling price, when we conclude this in to a small range of scales,

This type of plot will appear,

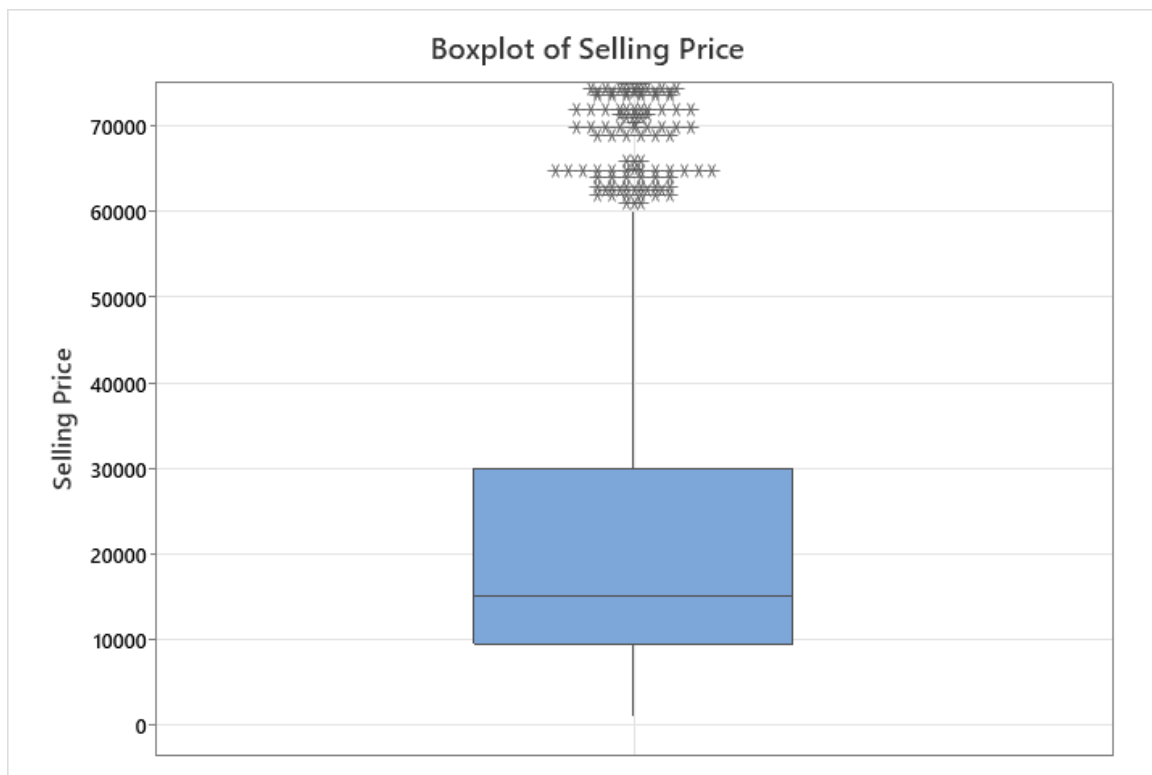


Figure 21 Boxplot of selling Price with reduce scale

## Original Price

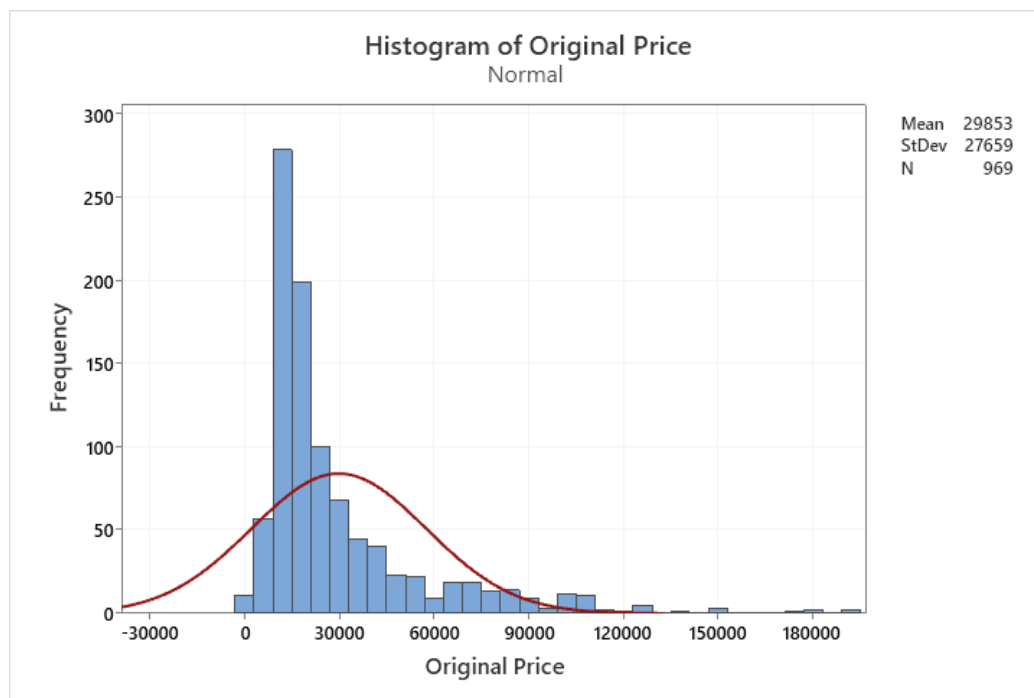


Figure 22 Histogram of Original Price

We can see this Distribution is approximately positively skewed. And it is Unimodal. We can see only one peak.

## Statistics

| Variable       | N   | N*   | Mean  | SE Mean | StDev | Minimum | Q1    | Median | Q3    | Maximum |
|----------------|-----|------|-------|---------|-------|---------|-------|--------|-------|---------|
| Original Price | 969 | 1678 | 29853 | 889     | 27659 | 1599    | 12999 | 18999  | 34999 | 189999  |

Figure 23 Descriptive Analysis of Original Price

- And There are outliers. (According to IQR \*1.5 rule).
- The data is centered around selling price RS.18999.



- These all-original price between 0 – 189999 rupees.
- Most brands have original prices between 9000 – 15000 rupees.
- We can see what the maximum and minimum original prices are by looking above bar chart.



Figure 24 Boxplot of Original Price

As we can see there are lot of outliers for the box plot of original price, when we conclude this into a small range of scales,

This type of plot will appear,



Figure 25 Boxplot with original price with reduce scale

## Storage

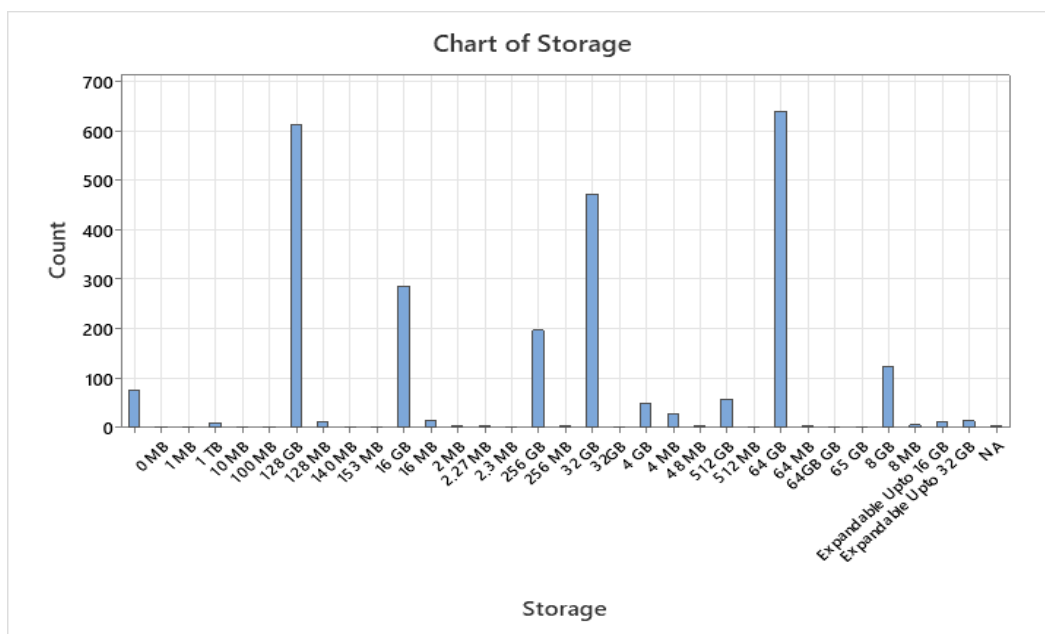


Figure 26 Bar chart of storage

- The "Chart of Storage" bar graph shows the frequency of occurrences for different storage capacity, most likely from a dataset related to electronics.
- The x-axis shows various storage capacity, including some extendable storage options, ranging from as little as 0MB to 64GB.
- The count is displayed on the y-axis, with the highest counts ever exceeding 700.
- The dataset's most popular storage capacities are 128MB, 32GB, and 64MB; 128MB appears most frequently, followed by 32GB.
- Expandable storage, 1MB, and 1GB are examples of less-used storage solutions with much lower counts, indicating their limited availability or utilization.

## Memory

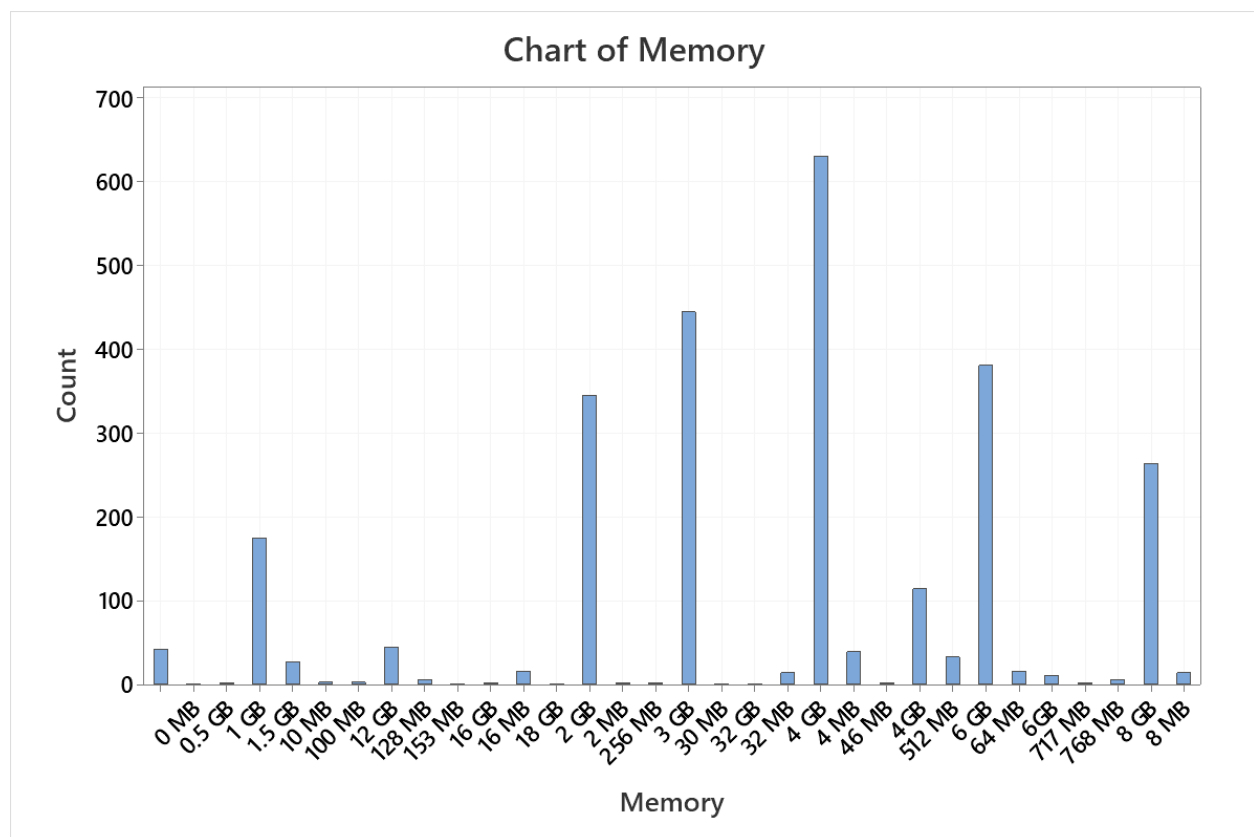


Figure 27 Bar chart of Memory

- The "Chart of Memory" bar graph shows how frequently certain memory sizes occur in a dataset.
- The y-axis displays the number of occurrences for each memory size, while the x-axis depicts different memory capacities, ranging from as little as 0MB to 8GB.
- The two memory sizes that are most commonly utilized are 4MB and 2GB, with 4MB being the most popular.
- Although they are utilized less frequently, other memory sizes like 1GB, 3GB, 64MB, and 8MB also show up with considerable counts.
- The graph shows that some memory sizes are much more popular than others, with extremely few or no occurrences for various lesser and bigger capacities.

## **Bivariate analysis**

Bivariate analysis is the simultaneous analysis of two variables (attributes). It explores the concept of relationship between two variables, whether there exists an association and the strength of this association, or whether there are differences between two variables and the significance of these differences. There are three types of bivariate analysis.

- Categorical Data – Categorical Data
- Numerical Data – Categorical Data
- Numerical Data – Numerical Data

## Original price and selling price



Figure 28 Boxplot of Original Price and Selling Price

### Statistics

| Variable       | N    | N*   | Mean  | SE Mean | StDev | Minimum | Q1    | Median | Q3    | Maximum |
|----------------|------|------|-------|---------|-------|---------|-------|--------|-------|---------|
| Original Price | 969  | 1678 | 29853 | 889     | 27659 | 1599    | 12999 | 18999  | 34999 | 189999  |
| Selling Price  | 2644 | 3    | 26461 | 580     | 29833 | 1000    | 9490  | 14999  | 29999 | 179900  |

Figure 29 Descriptive Analysis of Original price and selling price

- The boxplot shows the distribution of original prices and selling prices for a dataset.
- Both prices have a similar spread, with most values falling in a range between 25,000 and 75,000.
- There are several data points (shown as stars) that are much higher than the rest, indicating outliers. These outliers reach up to around 200,000.

- The central line inside each box shows the median price, which divides the data into two equal halves.
- The original and selling prices show a very steady overall price distribution.



Figure 30 Scatter Plot of Original Price and selling Price

- This scatter plot displays the relationship between original prices and selling prices. As the original price increases, the selling price generally increases as well.
- Most data points are grouped along the diagonal line, showing that many items have similar original and selling prices.

## Selling price and Brand

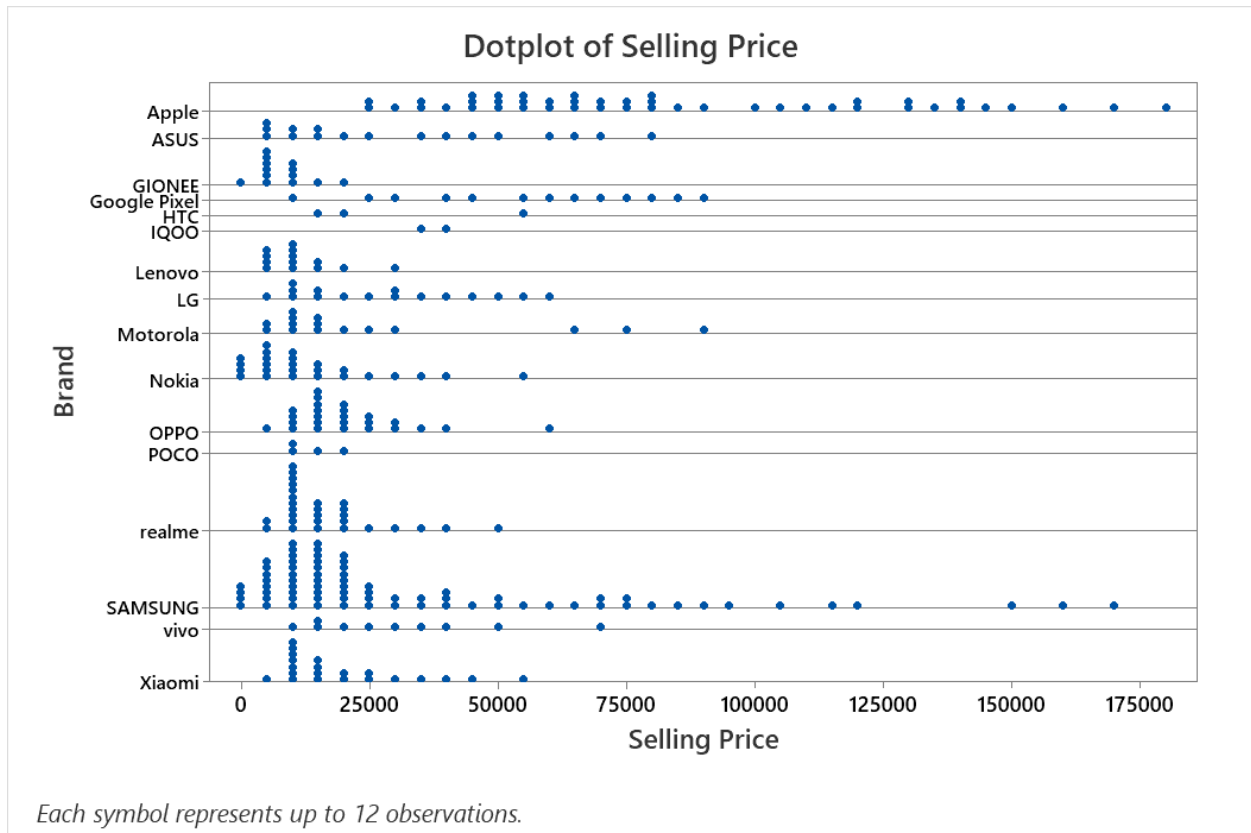


Figure 31 Dot plot of selling price and Brand

- The distribution of selling prices for several smartphone brands is displayed in this dot plot.
- Up to 12 devices at a specific price range are represented by each dot. While companies like Apple and Samsung offer a wider variety of costs, including numerous high-end devices priced above 50,000 and even up to 175,000, most brands, like Nokia, Realme, and Xiaomi, have devices that are mainly in the lower price range (around 25,000).
- This suggests that although some companies, like Apple and Samsung, provide high-end devices, others concentrate more on economical options.

## Rating and Brand

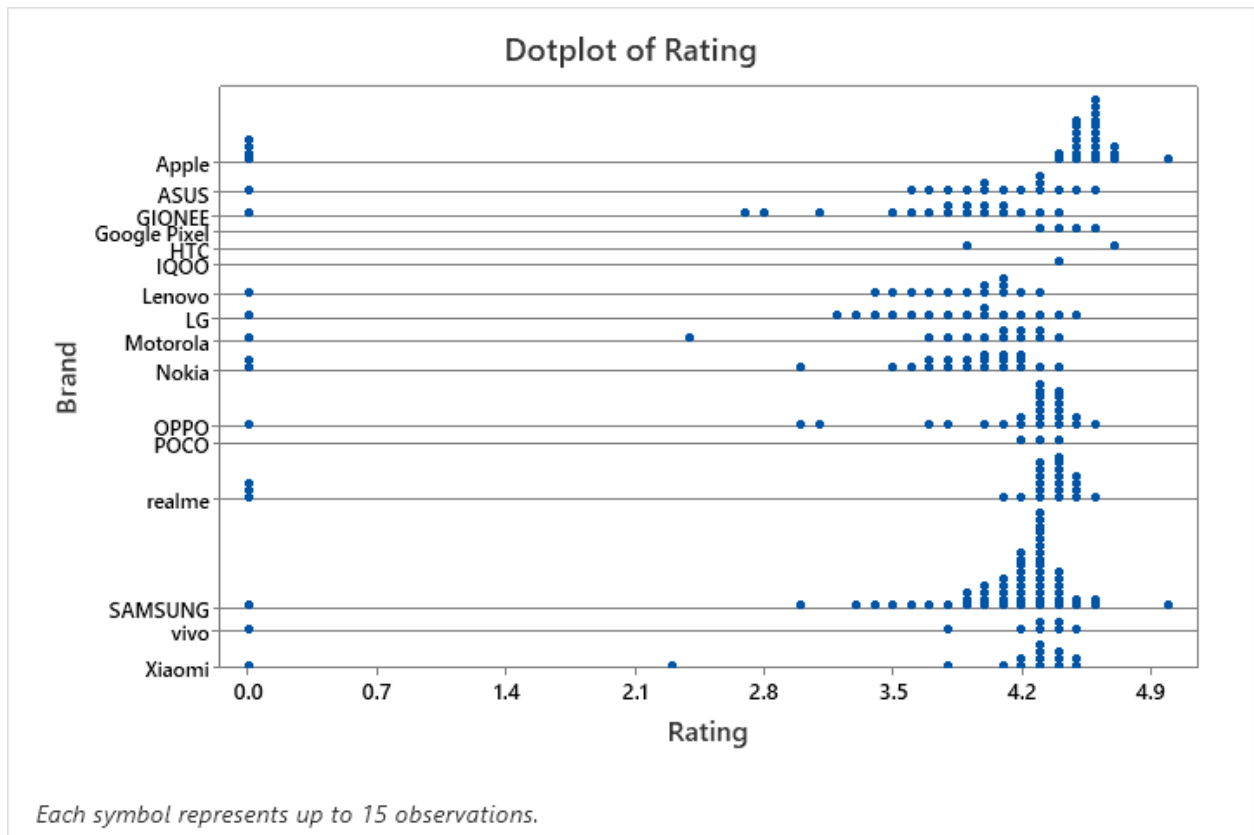


Figure 32 Dot plot of Rating and Brand

- The distribution of ratings for different smartphone brands is shown in this dot plot, where each dot represents up to 15 observations.
- The smartphone brands are listed on the y-axis, and the rating scale, which runs from 0 to 5, is displayed on the x-axis.
- Reviews for most products are typically positive, with ratings centered around higher values, especially between 4.0 and 4.9. Some brands, like ASUS, GIONEE, and Lenovo, display a wider range of ratings, including lower values, while brands like Apple, Samsung, and Xiaomi display dense clusters on the higher end of the scale.



- There are fewer positive evaluations for brands like Lenovo, LG, and Nokia, and some judgments are below 3.0. The plot illustrates, in general, how certain brands often score higher while receiving more varied input from consumers.

## Original Price and Brand

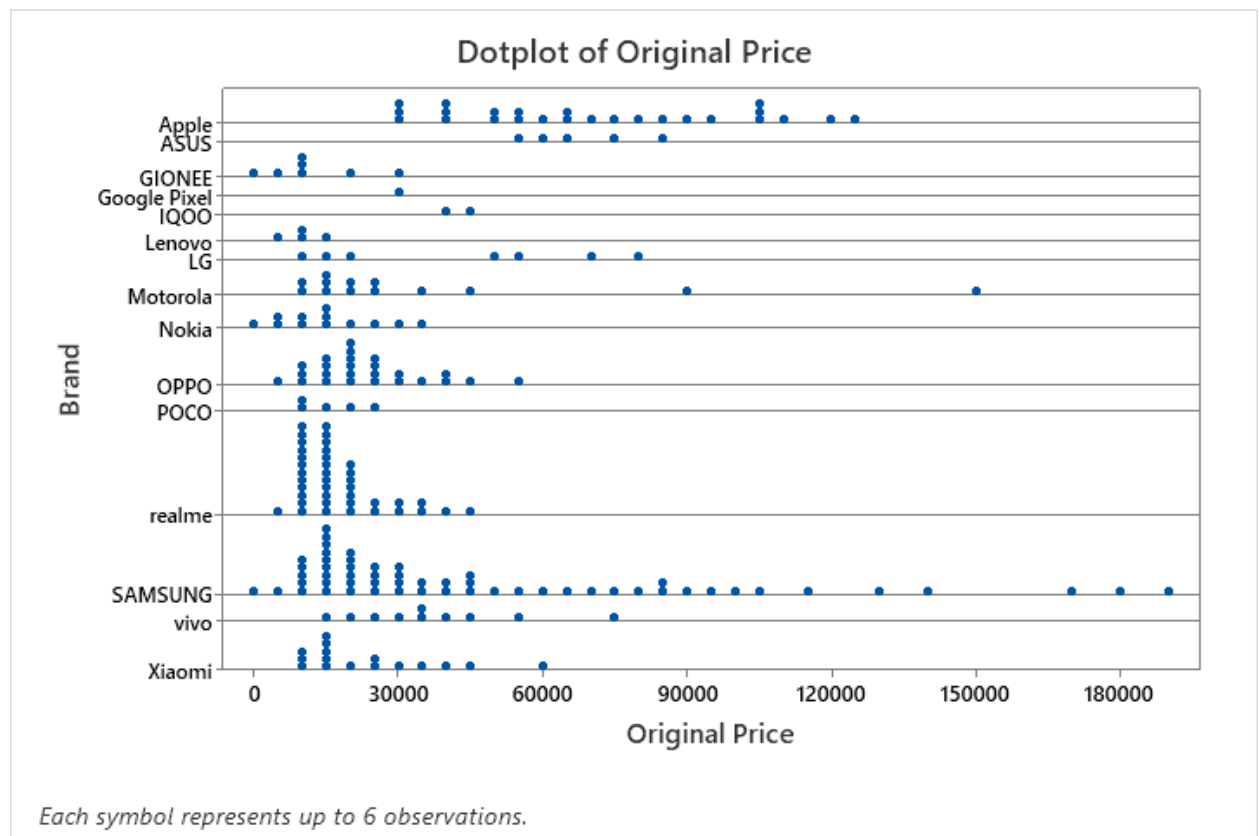


Figure 33 Dot plot of Original Price and Brand

- The distribution of the initial costs for different smartphone brands is seen in this dot plot. Up to six observations are shown by each dot on the plot.
- The initial pricing (in units) is shown along the horizontal axis, while the brands are listed on the vertical axis.

- The plot demonstrates the broad range of pricing offered by companies such as Apple, Samsung, and Google Pixel, with some models costing up to or exceeding one hundred thousand units.
- Brands like as ASUS, Xiaomi, and OPPO, on the other hand, usually concentrate at lower price points, usually under 40,000 units. The plot demonstrates how some companies, like Samsung and Apple, provide high-end devices at significantly higher price points than others.

## Correlation

### Selling Price and Original Price

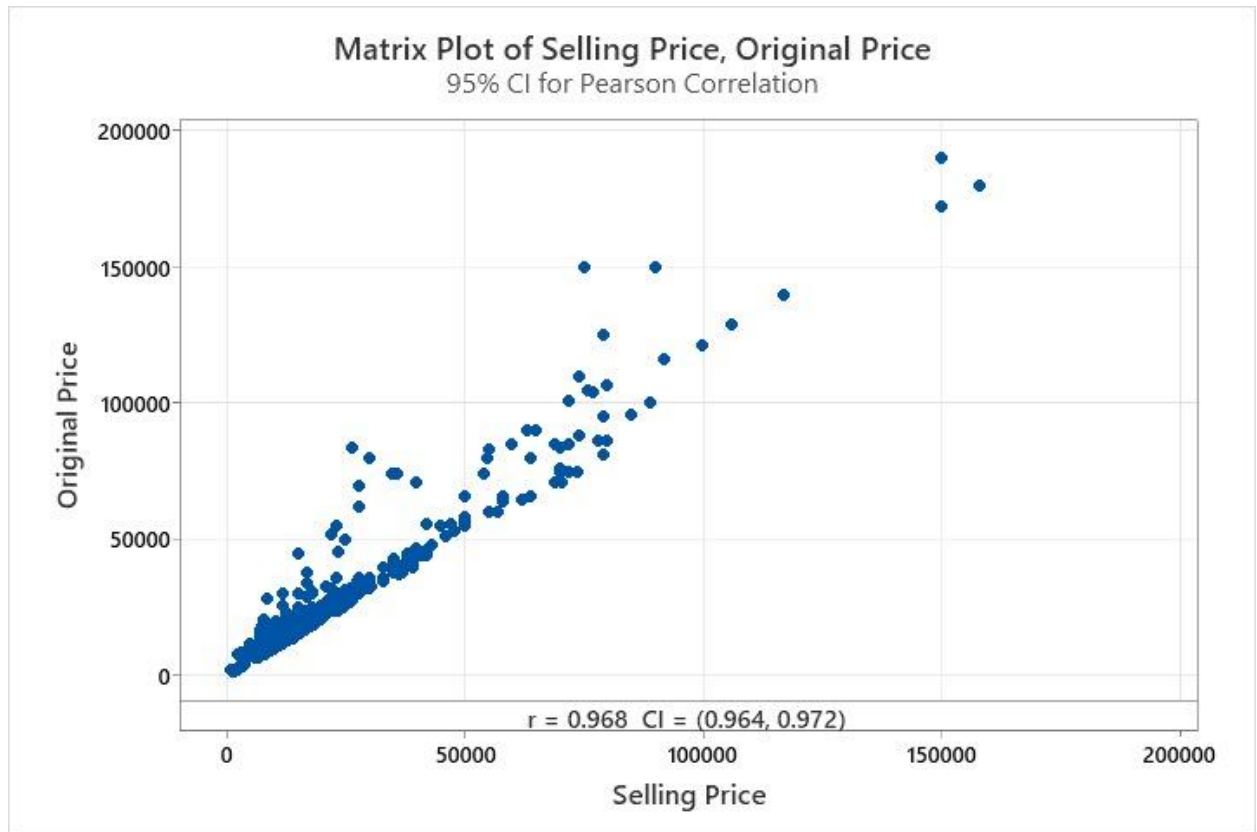


Figure 34 Matrix plot of Selling Price and Original Price

#### Method

Correlation type      Pearson  
Number of rows used   969

#### Correlations

|                | Selling Price |
|----------------|---------------|
| Original Price | 0.968         |

Figure 35 Correlation of Selling price Vs Original Price

- ❖ This matrix plot shows the relationship between the selling price and original Price.
- ❖ The correlation coefficient is 0.968.
- ❖ Therefore, the association between selling price and original Price can be seen as strong positive linear association.

## Ratings and selling price

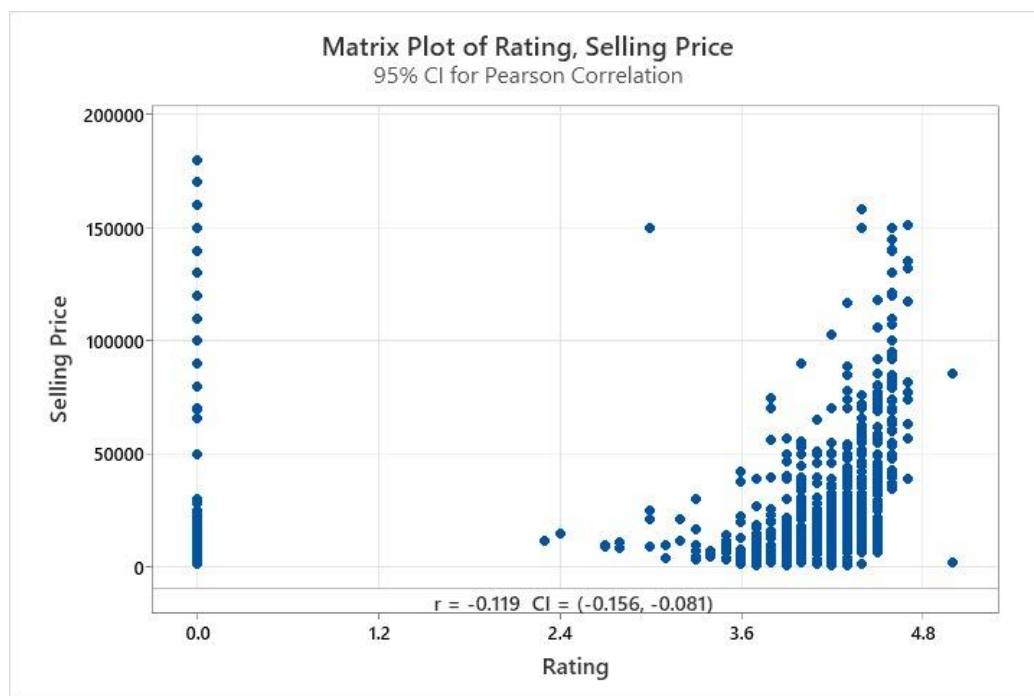


Figure 36 Matrix plot of Rating and selling Price

### Method

Correlation type      Pearson  
Number of rows used   2644

### Correlations

|               | Rating |
|---------------|--------|
| Selling Price | -0.119 |

Figure 37 Correlation of Rating Vs Selling Price

- ❖ This matrix plot shows the relationship between the selling price and Rating.
- ❖ The correlation coefficient is -0.119.
- ❖ Therefore, the association between selling price and Rating can be seen as a weakly negative linear association.

## Ratings and Original Price

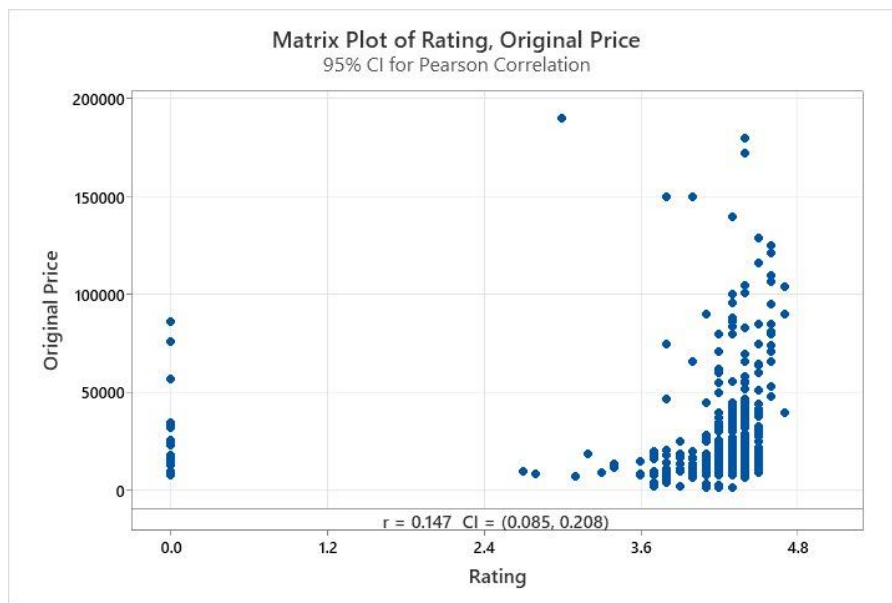


Figure 38 Matrix plot of Rating and Original Price

### Method

Correlation type      Pearson  
Number of rows used 969

### Correlations

|                | Rating |
|----------------|--------|
| Original Price | 0.147  |

Figure 39 Correlation of Rating Vs Original Price

- ❖ This matrix plot shows the relationship between the Rating and original Price.
- ❖ The correlation coefficient is 0.147.
- ❖ Therefore, the association between Ratings and original Price can be seen as weakly positive linear association.

## Regression

### Selling Price and Original Price



Figure 40 Plot of Original Price Vs selling Price with Fitted Line

Error Variance Ratio (Original Price/Selling Price): 1

Regression Equation  
Original Price = - 1181 + 1.265 Selling Price

#### Coefficients

| Predictor     | Coef        | SE Coef | Z        | P     | Approx 95% CI        |
|---------------|-------------|---------|----------|-------|----------------------|
| Constant      | -1181.04131 | 343.575 | -3.4375  | 0.001 | (-1854.44, -507.646) |
| Selling Price | 1.26534     | 0.011   | 119.8939 | 0.000 | (1.24, 1.286)        |

#### Error Variances

| Variable       | Variance |
|----------------|----------|
| Original Price | 18996280 |
| Selling Price  | 18996280 |

Figure 41 Numerical Measures of selling price Vs original price fitted line

The graph shows a scatter plot of "Original Price" versus "Selling Price," along with a fitted regression line. Here's a detailed description and analysis.

#### Scatter Plot Description:

##### 1. Axes:

- ❖ The x-axis represents the "Selling Price."
- ❖ The y-axis represents the "Original Price."

##### 2. Data Points:

- ❖ The scatter plot shows a positive linear relationship between Selling Price and Original Price. As the Selling Price increases, the Original Price tends to increase as well.
- ❖ The data points are closely clustered around the fitted regression line, indicating a strong linear association between the two variables.

##### 3. Fitted Regression Line:

- ❖ The red line represents the fitted regression line based on the regression equation:

$$\text{Original Price} = -1181 + 1.265 \times \text{Selling Price}$$

- ❖ This line predicts the Original Price based on the Selling Price and shows the general trend of the data.

#### Regression Analysis:

##### 1. Regression Equation:

- ❖ The regression equation,  $\text{Original Price} = -1181 + 1.265 \times \text{Selling Price}$  suggests that for every unit increase in the Selling Price, the Original Price increases by approximately 1.265 units.



## 2. Error Variance Ratio:

- ❖ The error variance ratio (Original Price/Selling Price) is (1), indicating that the variances of the errors in the model are equal for both variables.

## 3. Strength of Association:

- ❖ The close clustering of data points around the regression line implies a strong linear relationship between the Selling Price and the Original Price. The positive slope of the line (1.265) indicates that higher Selling Prices are associated with higher Original Prices.

## 4. Outliers:

- ❖ There are a few points that deviate more from the regression line, particularly at higher Selling Prices, which might be outliers or indicate some variation not captured perfectly by the model.

## Overall Interpretation:

### ○ Positive Linear Relationship:

The scatter plot with the fitted regression line clearly shows a positive linear relationship between Selling Price and Original Price. As the Selling Price increases, the Original Price also increases.

### ○ Model Fit:

The regression model fits the data well, as evidenced by the tight clustering of the data points around the regression line.

- Practical Implication:

The regression analysis suggests that Selling Price is a strong predictor of Original Price. The model can be used to estimate the Original Price based on the Selling Price with reasonable accuracy, especially within the range of data presented.

This analysis strengthens the positive association between the two variables and the effectiveness of the linear regression model in capturing this relationship.

## Original Price and Rating



Figure 42 Plot of Rating Vs Original Price with fitted line

Error Variance Ratio (Rating/Original Price): 1

Regression Equation

Rating = 3.920 + 0.000 Original Price

### Coefficients

| Predictor      | Coef    | SE Coef   | Z       | P     | Approx 95% CI       |
|----------------|---------|-----------|---------|-------|---------------------|
| Constant       | 3.92050 | 0.0456972 | 85.7929 | 0.000 | (3.83093, 4.01006)  |
| Original Price | 0.00001 | 0.0000011 | 4.6335  | 0.000 | (0.000000, 0.00001) |

### Error Variances

| Variable       | Variance |
|----------------|----------|
| Rating         | 0.933201 |
| Original Price | 0.933201 |

Figure 43 Numerical Measures of Original price Vs Rating fitted line

The image shows a scatter plot of "Rating" versus "Original Price" along with a fitted regression line. Below is a detailed description and analysis:

#### Scatter Plot Description:

##### 1. Axes:

- ❖ The x-axis represents the "Original Price."
- ❖ The y-axis represents the "Rating."

## 2.Data Points:

- ❖ The scatter plot shows a weak positive linear relationship between the Original Price and Rating. The majority of the data points cluster around a rating between 3.5 and 5.0, with a wide range of Original Prices, but with little variation in the Rating.

## 3. Fitted Regression Line:

- ❖ The red line represents the fitted regression line based on the regression equation:

$$\text{Rating} = 3.920 + 0.0001 \times \text{Original Price}$$

- ❖ The nearly flat line suggests that the Original Price has almost no effect on the Rating, as indicated by the coefficient of 0.0001.

## Regression Analysis:

### 1. Regression Equation:

The regression equation  $\text{Rating} = 3.920 + 0.0001 \times \text{Original Price}$  indicates that the slope is nearly zero, meaning the Original Price does not have a significant impact on the Rating.

### 2.Error Variance Ratio:

The error variance ratio (Rating/Original Price) is (1), indicating equal variances for the errors associated with the Rating and Original Price.

### 3. Strength of Association:

The scatter plot shows that Ratings are relatively constant, irrespective of the Original Price. This is further reinforced by the flat regression line, which implies no significant association between the variables.

#### 4. Outliers:

A few outliers are visible at higher Original Prices with varying Ratings, but they do not significantly impact the overall relationship.

#### Overall Interpretation:

- Weak or No Relationship:

The regression analysis and scatter plot suggest that there is no significant relationship between the Original Price and the Rating. The coefficient for the Original Price is zero, indicating that changes in Original Price do not predict changes in Rating.

- Model Fit:

The model does not fit the data well in terms of predicting Rating based on Original Price, as the Ratings are relatively consistent across different price points.

- Practical Implication:

Since the Original Price does not appear to influence the Rating, other factors may be more important in determining Ratings. This suggests that price alone does not affect the perceived quality or satisfaction represented by the Rating.

This analysis highlights the lack of correlation between the two variables, with the regression model confirming that the Original Price is not a significant predictor of Rating.

## Discussion

In this analysis report, we have used number of variables to carry out an accurate report. Univariate analysis is used to get better idea on each variable and bivariate analysis is used to highlight the relationships between two variables. Also, we used correlation and regression to be measuring this relationship numerically.

By using univariate and bivariate analysis, along with correlation and regression techniques, we can gain a comprehensive understanding of how phone specifications influence pricing and consumer behavior. These methods allow us to numerically quantify relationships between different variables and provide actionable insights for both the supply and demand sides of the mobile phone market.

## Conclusion

At the end of this analysis, the following are many factors we observed.

In conclusion, the analysis of phone brand specifications reveals significant relationships between technical features such as screen size, battery capacity, camera quality, processor performance, and storage capacity, and the market positioning of different phone brands. Premium brands tend to focus on high-end specifications that drive up prices, appealing to consumers who prioritize performance, design, and cutting-edge technology. Conversely, budget brands offer a balanced mix of specifications at lower price points, catering to more cost-conscious consumers.

The relationship between specifications and price is a key driver of consumer behavior, with higher-end features often justifying higher price tags. Additionally, brand reputation and consumer loyalty are closely tied to the consistent delivery of specific features, such as long battery life or superior camera performance. Brands that successfully align their specifications with consumer preferences can build strong market positions and foster long-term loyalty.

Overall, this report highlights the importance of understanding how phone specifications impact pricing, consumer choices, and brand perception in the competitive smartphone industry.



Figure 44 Conclusion

## References

- ❖ <https://www.kaggle.com/datasets/devsubhash/flipkart-mobiles-dataset>
- ❖ <https://www.minitab.com/en-us/support/>

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